

1. Demand for Money (i) Fisher — Transaction Approach (1911)

- $MV + M'V = PT \rightarrow Md = PT / V$
- **Flow** concept · V aur T ko **constant** maana · money sirf **medium of exchange**
- **Interest rate ka koi role nahi** — yahi iski sabse badi khaami

(ii) Cambridge — Cash Balance Approach

- $M = kPY$, jahan $k = 1/V$ (Cambridge k, Fisher ke V ka reciprocal)
- **Stock** concept · micro / behavioural · money = exchange + **store of value**
- Yahi se aage **Keynesian Liquidity Preference** develop hui

(iii) Keynes — Liquidity Preference Theory (1936)

- $Md = L_1(Y) + L_2(r)$
- **Transaction + Precautionary** → income (Y) ka function
- **Speculative** → interest rate (r) ka function, **inverse** · bond price aur r inversely related
- **Liquidity trap**: bahut low r par LP curve **perfectly elastic (horizontal)** → monetary policy fail → Keynes fiscal policy ke favour mein

(iv) Friedman — Modern Quantity Theory (1956)

- $Md/P = f(Y_p, w, r_m, r_b, r_e, \text{expected inflation, } u)$
- Money = **durable asset** · main determinant = **permanent income (Y_p)**, measured income nahi
- Income elasticity > 1 → money ek **luxury good**
- Md **stable** aur **interest-inelastic** → monetary policy **powerful** (Keynes ke bilkul ulta)

(v) Baumol — Inventory Theoretic Approach (1952)

- $C = \sqrt{(2bT / i)}$ · **Average cash balance = $C/2 = \sqrt{(bT / 2i)}$**
- **Transaction demand** ko interest-elastic banana (Keynes ne ise sirf income ka function maana tha)
- **Interest elasticity = $-1/2$ | Income elasticity = $+1/2$** → **economies of scale** in cash holding

(vi) Tobin — Portfolio / Risk-Aversion Approach (1958)

- Paper: "Liquidity Preference as Behaviour Towards Risk"
- **Speculative demand** ko risk-aversion se explain kiya — Keynes ka **all-or-nothing** hataya
- Money = zero return, zero risk | Bond = higher return, but risk → **diversified portfolio**
- 3 types: **Risk-averters** (majority), Plunger (risk-lover), Risk-neutral

★ Master Comparison Table

Basis	Fisher	Cambridge	Keynes	Friedman	Baumol	Tobin
Approach	Transaction (flow)	Cash balance (stock)	Liquidity preference	Wealth / asset theory	Inventory	Portfolio
Main determinant	P, T	k, P, Y	Y and r	Permanent income	T, b, i	Risk-return trade-off
Interest rate	No role	Implicit	Central	Negligible	Yes	Yes
Kya explain kiya	Transaction	Transaction	All 3 motives	Total (asset)	Transaction	Speculative
Md stable?	Stable	Stable	Unstable	Highly stable	—	—

⚠ Sabse Common Galti

Baumol = TRANSACTION demand ko interest-elastic banana | **Tobin = SPECULATIVE** demand ko risk-based banana. Exam mein ye ulta pooch ke phasaate hai.

2. Supply of Money

- Money supply = **stock** concept · sirf **public ke paas** wala money → banks, govt aur RBI ka cash **exclude**

Monetary Aggregates (Second Working Group, 1977)

Measure	Formula	Naam
M1	Currency with public + Demand Deposits + Other Deposits with RBI	Narrow Money
M2	M1 + Post Office Savings Deposits	—
M3	M1 + Time Deposits of banks	Broad Money = AMR main policy indicator
M4	M3 + Total Post Office deposits (NSC chhod ke)	Broadest

Liquidity order: **M1 > M2 > M3 > M4** · 1998 Y.V. Reddy Group → NM1-NM3, L1-L3

High-Powered Money (H)

- **H = C + R** (currency with public + bank reserves) — "Reserve Money / Base Money"
- **H = RBI ki liability** | **M = RBI + commercial banks** milkar banate hai
- Sources: Net RBI credit to Govt + credit to commercial sector + net forex assets + govt currency liabilities – net non-monetary liabilities

Money Multiplier

- **M = m × H** · **m = (1 + c) / (c + r)** · simple deposit multiplier **= 1/r**
- $c = C/D$ (currency-deposit ratio), $r = R/D$ (reserve-deposit ratio)
- m hamesha > 1 kyunki $(c + r) < (1 + c)$ → yahi **credit creation** hai

Determinants – Ek Line Mein

M badhta hai jab → **H ↑, r ↓, c ↓**

- **c badhta hai**: festival, election, black money, unbanked areas → M ghatta hai
- **c ghatta hai**: Jan Dhan, UPI, banking habit, financial deepening → M badhta hai
- **r badhta hai**: CRR/SLR ↑, repo ↑, low business confidence → M ghatta hai

Debate: Exogenous (Monetarist — RBI H control karta hai) vs Endogenous (Post-Keynesian — "loans create deposits")

3. Money Market & Changing Role of RBI

Basis	Money Market	Capital Market
Maturity	1 din – 1 saal	1 saal se zyada
Regulator	RBI	SEBI
Purpose	Working capital	Fixed capital

Structure: Organised (RBI, banks, DFHI, PDs, MFs) + **Unorganised** (indigenous bankers — Shroffs, Multanis, Chettiars; moneylenders, chit funds, nidhis) — unorganised sector RBI ke control se **bahar** hai.

Instruments

Instrument	Issuer	Maturity	Feature
Call Money	Banks (inter-bank)	1 din (overnight)	Call rate — sabse volatile
Notice / Term Money	Banks	2–14 din / 15 din–1 saal	—
Treasury Bills	Central Govt (RBI issue)	91, 182, 364 din	Zero coupon , discount pe issue
CMB	Govt of India (2010)	91 din se kam	Temporary cash mismatch
Commercial Paper	Corporates, PDs, FIs (1990)	7 din – 1 saal	Unsecured, min ₹5 lakh
Certificate of Deposit	Banks & FIs (1989)	7 din – 1 saal	Negotiable, min ₹5 lakh
CBLO → TREPS	CCIL	Overnight+	CBLO 2018 mein replace hua

Trap: **CP = company** issue karti hai | **CD = bank** issue karta hai.

Defects & Reforms

- **Defects**: **dichotomy** (organised + unorganised), no integration, diverse interest rates, seasonal stringency, underdeveloped bill market, volatile call rate
- **Nagpal Committee (1977)** = money market reform ka blueprint → DFHI (1988), CP & CD (1989-90), MMMF (1992), PDs (1995), **LAF (2000)**, CCIL (2001)

RBI — Changing Role (Phases)

- **Est. 1935** (Hilton Young Commission), **nationalised 1949** · **₹1 note min value = Ministry of Finance**, baaki notes RBI · **Minimum Reserve System 1957** — ₹200 cr (₹115 cr gold + ₹85 cr forex)
- **Pre-1991**: direct controls, **CRR 15% + SLR 38.5%**, administered rates, ad-hoc T-bills se automatic monetisation

- Post-1991: **1997-98 T-bill auctioned 1997** → WMA; FRBM 2003; CRR-SLR cut; interest deregulation; prudential norms
- Post-2016 (FIT regime): Urjit Patel Committee (2014) → RBI Act amendment 2016 → **MPC = 6 members** (3 RBI + 3 Govt), Governor ka casting vote; target **CPI 4% ± 2%**
- New roles: **SDF (April 2022)** = corridor floor, CBDC / Digital Rupee (2022), UPI, PCA framework, NBFC scale-based regulation

4. Inflation

Inflation = general price level mein **sustained & continuous rise**, jisse money ki **purchasing power girti** hai. (Ek baar ka price rise inflation nahi — "sustained" key word hai.)

Zaroori Terms

Disinflation	Inflation rate ka ghatna — prices abhi bhi badh rahe, bas dheere (8% → 5%)
Deflation	Price level mein sustained fall
Reflation	Deflation ke baad jaan-boojh kar price badhana (recovery ke liye)
Stagflation	Inflation + unemployment + low growth ek saath — Phillips Curve ko contradict karta hai (1970s oil shock)
Skewflation	Sirf kuch goods (jaise food) mein price rise — India-specific term
Core Inflation	Headline — food & fuel

Types by Cause

- **Demand-pull** — "too much money chasing too few goods" → **price ↑, output ↑**
Keynes: AD > AS at full employment = inflationary gap · Friedman: "Inflation is always and everywhere a monetary phenomenon"
- **Cost-push** — wage-push (wage-price spiral), profit-push, supply shock, tax-push → **price ↑ par output ↓** → **is se stagflation aata hai**
- **Structural** (developing economies — bottlenecks) · **Imported** (crude, edible oil)

Measurement in India

Basis	CPI	WPI
Agency	NSO, MoSPI	Office of Economic Adviser, DPIIT (Min. of Commerce)
Base year	2012	2011-12
Level	Retail	Wholesale / bulk
Services	Included	Excluded — sabse badi khaami
Highest weight	Food & Beverages ~45.9%	Manufactured 64.2% (Primary 22.6%, Fuel 13.2%)
Policy use	RBI ka inflation target anchor (2014 se)	Pehle headline index tha

- **GDP Deflator** = **(Nominal GDP / Real GDP) × 100** — sabse **comprehensive**, par annual/lagged
- **CPI-IW** — Labour Bureau, base 2016, **DA calculation** ke liye

Trends (Broad Picture)

1974-75 → **highest ever ~25%** (oil shock + drought) · 1991 → ~13-14% (BOP crisis) · 2008-13 → double-digit, food-driven · 2014-19 → moderation ~4% (FIT + low crude) · 2022 → **6% band breach** (COVID + Russia-Ukraine war), RBI ko Parliament ko report dena pada.

Consequences — Kaun Fayde Mein, Kaun Nuksan Mein

✓ GAINERS	✗ LOSERS
Debtors (loan ki real value girti)	Creditors / lenders
Producers / businessmen	Fixed income group — pensioners, salaried
Equity & share holders	Bond & debenture holders
Real asset holders (land, gold, property)	Savers (real interest rate negative)
Government (real debt ↓, "inflation tax")	Wage earners (wages prices se peeche)

- Inflation **regressive** hai — gareeb sabse zyada affected (food ka share zyada)
- Other costs: **menu cost**, **shoe-leather cost**, **bracket creep**, seigniorage, BOP deficit, currency depreciation
- **Fisher Effect**: **i = r + expected inflation** (nominal = real + expected)

Remedies

- **Monetary**: repo ↑, CRR / SLR ↑, **OMO** — **securities bech do**, MSS, margin requirement ↑, credit rationing, moral suasion
- **Fiscal**: public expenditure ↓, taxes ↑, public borrowing ↑, surplus budget, **deficit financing band karo**
- **Supply-side**: production ↑, imports, **buffer stock release**, PDS strengthen, anti-hoarding (Essential Commodities Act), export ban, wage-price freeze

★ Exam Logic (Yaad Rakho)

Demand-pull → monetary + fiscal measures kaam karte hai.

Cost-push → sirf **supply-side** measures. Yahan contractionary monetary policy ulta output aur gira degi.

5. Monetary Policy

- Objectives: **price stability (primary, post-2016)**, growth, exchange rate stability, full employment, financial stability, equitable credit
- Impossible Trinity** (Mundell-Fleming): fixed exchange rate + free capital movement + independent monetary policy — teeno ek saath nahi. India ne **middle path** chuna.

LAF Corridor

MSF (Ceiling) — REPO (Policy Rate) — SDF (Floor)

Corridor width normally 50 bps (± 25 bps) · SDF (April 2022) ne Reverse Repo ko floor ke roop mein replace kiya · Bank Rate = MSF

Instruments

Instrument	Kya hai	Inflation mein
Repo Rate	RBI banks ko short-term loan deta hai (G-Sec ke against) — main policy rate	↑
SDF	Banks RBI ke paas bina collateral paisa rakhein — corridor floor	↑
MSF	SLR securities ke against emergency overnight borrowing — ceiling	—
CRR	NDTL ka % RBI ke paas cash mein — koi interest nahi	↑
SLR	NDTL ka % bank khud ke paas liquid assets mein — interest milta hai	↑
OMO / MSS	G-Sec khareedna-bechna · bechna = liquidity absorb	Bechna

Qualitative: margin requirement, consumer credit regulation, credit rationing, **moral suasion**, direct action, PSL norms.

Trap: Quantitative = credit ka volume control | Qualitative = credit ki direction / allocation control.

Transmission & Limitations

- Chain: **Policy rate ↓ → lending rate ↓ → credit ↑ → investment & consumption ↑ → AD ↑**
- Rate evolution: PLR (1994) → BPLR (2003) → **Base Rate (2010)** → **MCLR (2016)** → **EBLR (Oct 2019)**
- Weak transmission kyu:** high NPAs, small savings ke administered rates, rigid deposit rates
- Limitations:** bada unorganised sector, black money, underdeveloped money market, cost-push inflation, liquidity trap, fiscal dominance

6. Financial Sector Reforms

Pre-1991 problem: **CRR 15% + SLR 38.5% = 53.5% funds pre-empted** → "**financial repression**", administered rates, directed credit, no prudential norms.

Narasimham Committee I (1991) — reform ka blueprint	Narasimham Committee II (1998) — strengthening
- SLR 38.5% → 25%, CRR reduce - Interest rate deregulation Prudential norms — income recognition, NPA classification, provisioning CAR 8% (Basel I) - Four-tier banking structure - Branch licensing abolish, new private banks ki entry - ARF, Banking Supervision Board, dual control khatam	CAR 8% → 10% Merger of strong banks; weak banks close/restructure Net NPA < 3% ARCs (Asset Reconstruction Companies) Narrow banking for weak banks - Recapitalisation of PSBs, govt stake 33% tak - Board for Financial Supervision strengthen

Milestones

DRT 1993 · New private banks 1993-94 (HDFC, ICICI, Axis, IndusInd) · SEBI statutory + NSE 1992 · Current account convertibility 1994 · BFS 1994 · FEMA 1999 · **SARFAESI 2002** (court ke bina recovery) · FRBM 2003 · Payments Banks & SFBs 2015-16 · **IBC 2016** · GST 2017 · Mega-merger 2019 (**PSBs 27 → 12**) · NARCL "bad bank" 2021 · CBDC 2022

Committees yaad rakho: Chakravarty (1985) monetary system · Vaghul (1987) money market · Narasimham (1991, 1998) banking · Tarapore (1997, 2006) capital account convertibility · Raghuram Rajan (2008) · Nachiket Mor (2013) financial inclusion · Urjit Patel (2014) monetary framework · P.J. Nayak (2014) bank boards

7. Commercial Banking Functions

Definition — BR Act 1949, Sec 5(b): public se **deposits accept** karna (lending/investment ke liye), jo **repayable on demand** ho aur cheque/draft/order se withdrawable ho.

Scheduled Bank = RBI Act ka 2nd Schedule, paid-up capital $\geq$ ₹5 lakh.

A. Primary Functions

- 1. **Accepting Deposits** — Current (**no interest**, overdraft) · Savings · Fixed/Term (**highest interest**) · Recurring
DICGC insurance = ₹5 lakh per depositor per bank (2020 se)
- 2. **Advancing Loans** — cash credit, overdraft, term loans, bill discounting, money at call
- 3. **Credit Creation** — banks ka **most unique function**. **Total deposits = Primary deposit × 1/r**
"Banks are not merely purveyors of money but manufacturers of money."
- 4. **Payment mechanism** — cheque system

B. Secondary Functions

- Agency**: cheque/bill collection, dividend-interest-pension collection, bill & premium payment, fund transfer (NEFT/RTGS/IMPS/UPI), securities purchase-sale, trustee & executor
- General Utility**: locker, **Letter of Credit & Bank Guarantee**, forex dealings, cards, traveller's cheques, underwriting, merchant banking, bancassurance

C. Developmental / Social Functions

- PSU = 40% of AMR** (agriculture 18%, weaker sections 12%) — RRB & SFB ke liye **75%**
- Financial inclusion (Jan Dhan, BC model), rural branch expansion, DBT & MUDRA implementation

Liabilities	Assets
Capital & Reserves Deposits (largest) Borrowings, other liabilities	Cash in hand & with RBI Money at call & short notice Investments (G-Sec — SLR) Loans & Advances (most profitable) Fixed assets

3 Conflicting Principles of Bank Management

Liquidity · Profitability · Safety (Solvency)

Liquidity aur profitability mein **trade-off** hai — cash rakho to profit nahi, loan do to liquidity nahi.

Differentiated Banks (2015-16, Nachiket Mor Committee): **Payments Bank** — deposit limit ₹2 lakh, loan/credit card nahi de sakte | **Small Finance Bank** — 75% PSL, 50% loans ≤ ₹25 lakh.

★ Formula Sheet — Ek Nazar Mein

Concept	Formula
Fisher (Transaction)	$MV + M'V' = PT$
Cambridge (Cash Balance)	$M = kPY$, $k = 1/V$
Keynes (Liquidity Pref.)	$Md = L_1(Y) + L_2(r)$
Baumol (Inventory)	$C = \sqrt{(2bT/i)}$, average = $C/2$, elasticity: $i = -1/2$, $T = +1/2$
High-Powered Money	$H = C + R$
Money Supply	$M = m \times H$
Money Multiplier	$m = (1 + c) / (c + r)$, $c = C/D$, $r = R/D$
Simple Deposit Multiplier	$1/r$
Credit Creation	Primary deposit × 1/r
GDP Deflator	$(\text{Nominal GDP} / \text{Real GDP}) \times 100$
Fisher Effect	$i = r + \text{expected inflation}$
M1 / M3	$M1 = \text{Currency} + \text{Demand Deposits} + \text{Other deposits with RBI}$ $M3 = M1 + \text{Time Deposits}$

⚠ Last-Minute Traps

- Baumol** = transaction demand, **Tobin** = speculative demand — ulta mat karna
- Baumol elasticities: interest $-1/2$, income $+1/2$
- $k = 1/V$ — Cambridge k aur Fisher velocity reciprocal
- Friedman ke liye money = **luxury good** (income elasticity > 1) aur Md **stable**; Keynes ke liye **unstable**
- M3 = Broad Money = AMR** — teeno ek hi cheez
- Money supply mein **banks ka cash NAHI** ginte
- CPI base 2012**, **WPI base 2011-12** — dono alag · $WPI = DPIIT$, $CPI = NSO$
- Disinflation ≠ Deflation**

9. **CRR** pe interest nahi, **SLR** pe milta hai
10. **₹1 note** = **Ministry of Finance**, baaki **RBI**
11. **MPC** = **6 members**, quorum 4, Governor ka casting vote · target **CPI 4% ± 2%**
12. Target "miss" = **3 consecutive quarters** band se bahar → Parliament ko report
13. **SDF** = **floor (April 2022)**, **MSF** = ceiling, Bank Rate = **MSF**
14. **Credit creation** = commercial banks ka unique function; **note issue** = RBI ka
15. India mein **CAR 9% + CCB 2.5%**; Basel III global norm 8%
16. Chronology: Vaghul **1987** → ad-hoc T-bills abolished **1997** → SARFAESI **2002** → MPC **2016**

– Day 05 Complete · Next: Unit 6, Public Finance –